PLSC 40601

Week 2: Bootstrapping, bagging, honesty.

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Bootstrapping.

Bootstrapping

- Another approach, typically used to estimate the variability of an estimate over random samples, is bootstrapping.

- If we knew the CDF of our population, we would be able to exactly determine the sampling variation of our estimate.

- While we do not, we can *suppose* that the empirical CDF produced by the data that we observe is identical to the population CDF.

- We can then just resample with replacement from our observed data, and see how much our estimates vary across resamples.

Bootstrapping for variability of the estimate due to random sampling

The bootstrapping procedure is:

- For b in $1 \ldots B$:
  1. Take a sample of size N *with replacement* from the observed data.
  2. Apply the estimating procedure on the bootstrap sample.

- To get an estimate of the standard error of the estimate, calculate the standard deviation over these many bootstrap estimates.

Bootstrapping

- How can we translate this method to estimate test error?

Bootstrapping for test error

The bootstrapping procedure is:

- For b in $1 \ldots B$:
  1. Take a sample of size N *with replacement* from the observed data.
  2. Apply the fitting procedure on the bootstrap sample to produce $\hat{f}^b$.

- For each unit i, find the error from all of the bootstrap samples that do *not* contain i.

$$\widehat{\mathrm{Err}}_{Boot} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{|C^{-i}|} \sum_{b \in C^{-i}} L(y_i, \hat{f}^b(x_i))$$

C^{-i} is the set of bootstrap samples that do not contain i, $|C^{-i}|$ is the size of this set.

Bootstrapping

- How does the bootstrapping procedure perform?
- We still have the problem of too-small sample size; on average, we only have $0.632 \times N$ unique observations in each boostrap sample.
- This means the bootstrap error *over* estimates the test error.
- Solution:

$$\widehat{\mathrm{Err}}_{0.632} = 0.368\overline{\mathrm{err}} + 0.632\widehat{\mathrm{Err}}_{Boot}$$

  where $\overline{\mathrm{err}}$ is the training error.
- This works...OK.
- Some alternatives in Hastie et al. (2009).

Bagging.

Bagging

- We can combine estimates across samples to get smoother, or better estimators.

- Bootstrap aggregating.

Bootstrap estimation for aggregated fit

- For b in $1 \ldots B$:
  1. Take a sample of size N *with replacement* from the observed data.
  2. Apply the fitting procedure on the bootstrap sample to produce $\hat{f}^b$.

- The bagging estimate is

$$\hat{f}_{bag}(x) = \frac{1}{B} \sum_{i=1}^{N} \hat{f}^b(x)$$

Bagging

- This is less interesting for something like a linear model, where $\hat{f}_{bag}(x) \to \hat{f}(x)$ as $B \to \infty$, since all of our observations are equally weighted in the sample, we'll reproduce the same thing, or possibly a worse version of it.

- This is more interesting with something "ragged" like regression trees, where different trees can give us different branching behavior that we can smooth over.

- If we're using a classifier, each model can get a "vote" for each x, and the class with the most votes wins.

- Or we can use averages of classifiers to produce probabilities, rather than just class predictions.

Bagging

- How many bootstrap replicates?
- 25? 50? See how your results change with more replicates.

Honesty.

Honesty

- Returning to (causal) inference…we might like to use these methods to get valid inference, potentially on causal targets.

An honest tree algorithm

1. Split the sample into two folds.
2. Use the first fold to learn splits of the tree.
3. Estimate response within leaves using the second fold.

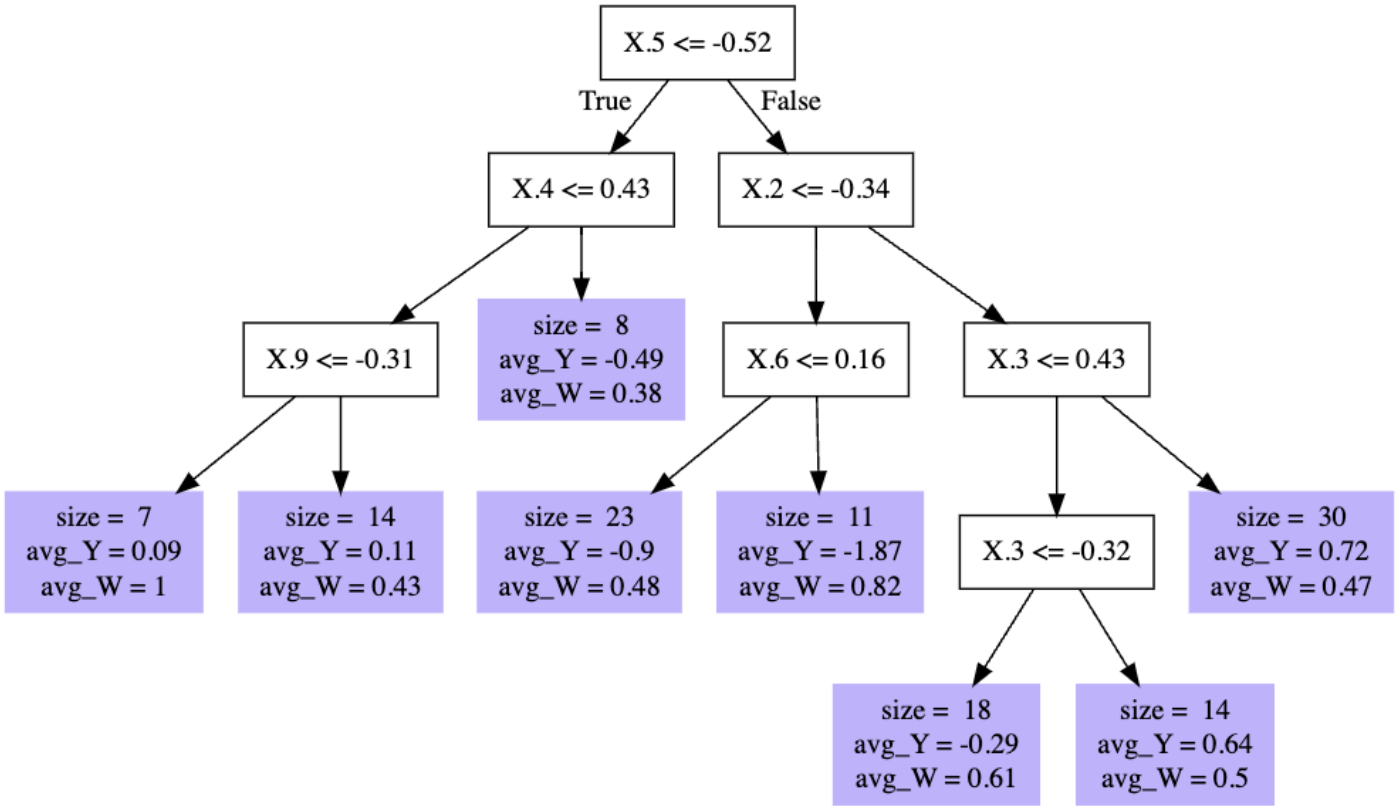
- This can result in some leaves being empty. Prune them?

- This procedure reduces bias relative to those proposed by Breiman (2001).

An honest tree algorithm

```
> library(grf)
> set.seed(60637)
> n <- 500
> p <- 10
> X <- matrix(rnorm(n * p), n, p)
> W <- rbinom(n, 1, 0.5)
> Y <- pmax(X[, 1], 0) * W + X[, 2] +
+   pmin(X[, 3], 0) + rnorm(n)
> c.forest <- causal_forest(X, Y, W)
```

An honest tree



References I

Breiman, L. (2001). Random forests. *Machine Learning*, 45:5–32.

Hastie, T., Tibshirani, R., and Friedman, J. H. (2009). *The elements of statistical learning: data mining, inference, and prediction*, volume 2. Springer.